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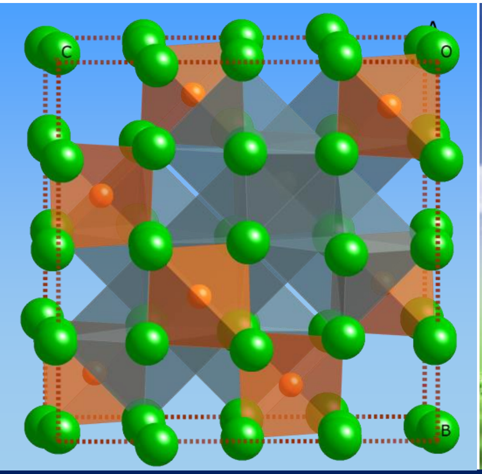
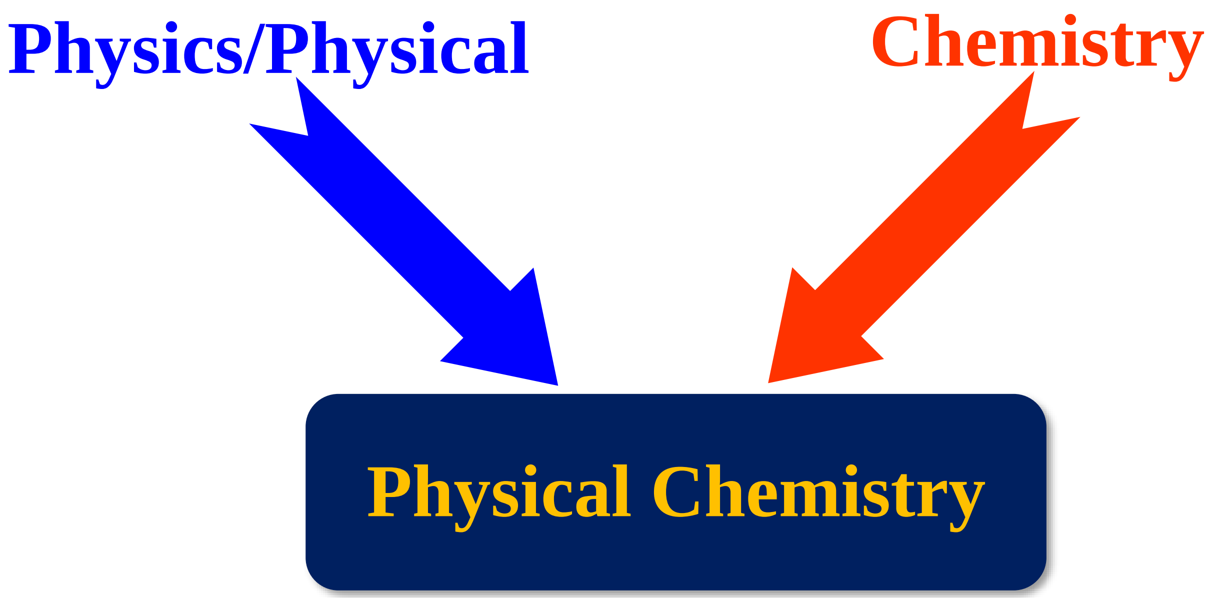


Physical Chemistry
物理化学

Dr. Leilei Tian(田雷蕾)



Physical theories and experimental methods

Nature and laws of chemical changes

研究所有物质体系的化学行为的原理、规律和方法的学科

Theoretical Foundation

- Thermodynamics
- statistics
- quantum mechanics

Methods

- Experimental methods
- Mathematics

Problems

- Direction, extent, energy –Thermodynamics
  - 化学变化的方向和限度问题
- Rate & mechanism - Dynamics
  - 化学反应的速率和机理问题
- Properties and structure –Structure
  - 物质结构和性能之间关系

Chapter 1: The properties of gases

Chapter 2: The first law of thermodynamics

Chapter 3: The second law of thermodynamics

Chapter 4: Physical Transformations of Pure Substances

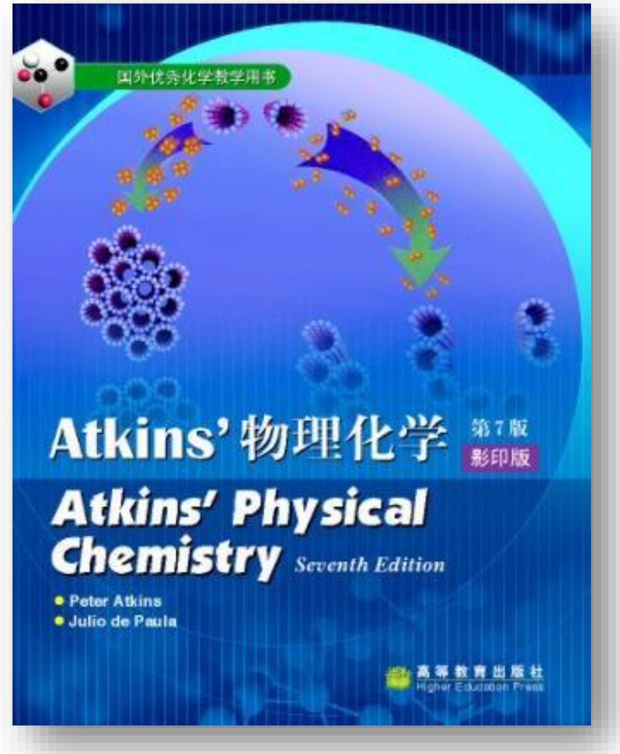
Chapter 5: Simple mixture

Chapter 6: Phase diagram

Chapter 7: Chemical equilibrium

Chapter 8: Electrochemistry equilibrium

P.W. Atkins, “Physical Chemistry”, Seventh Edition (or latest), Oxford University Press, 2003.



傅献彩、沈文霞、姚天扬等. 物理化学（上、下册）(第五版). 北京 高等教育出版社, 2006.

Physical Chemistry Class

- Tuesday (单、双) 10:20-12:10, Teaching Building, Room 503
- Thursday (双) 8:00-9:50, Teaching Building, Room 503

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Score

Attendance and quiz: 10%

Assignments: 15%
Every week

Middle Exam: 30%
Paper test

Final Exam: 45%
Paper test

群名称:2022春季-物理化学-田雷蕾
群 号:283257311

ETHICS: NO CHEATING AND PLAGIARISM

All these improper behaviors will cause you to fail the exam!

No excuses!

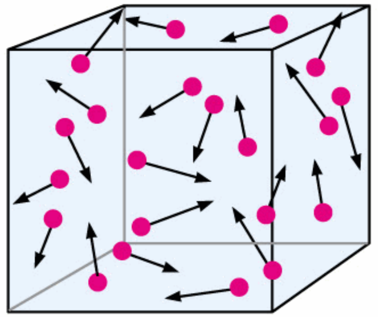
Chapter 1

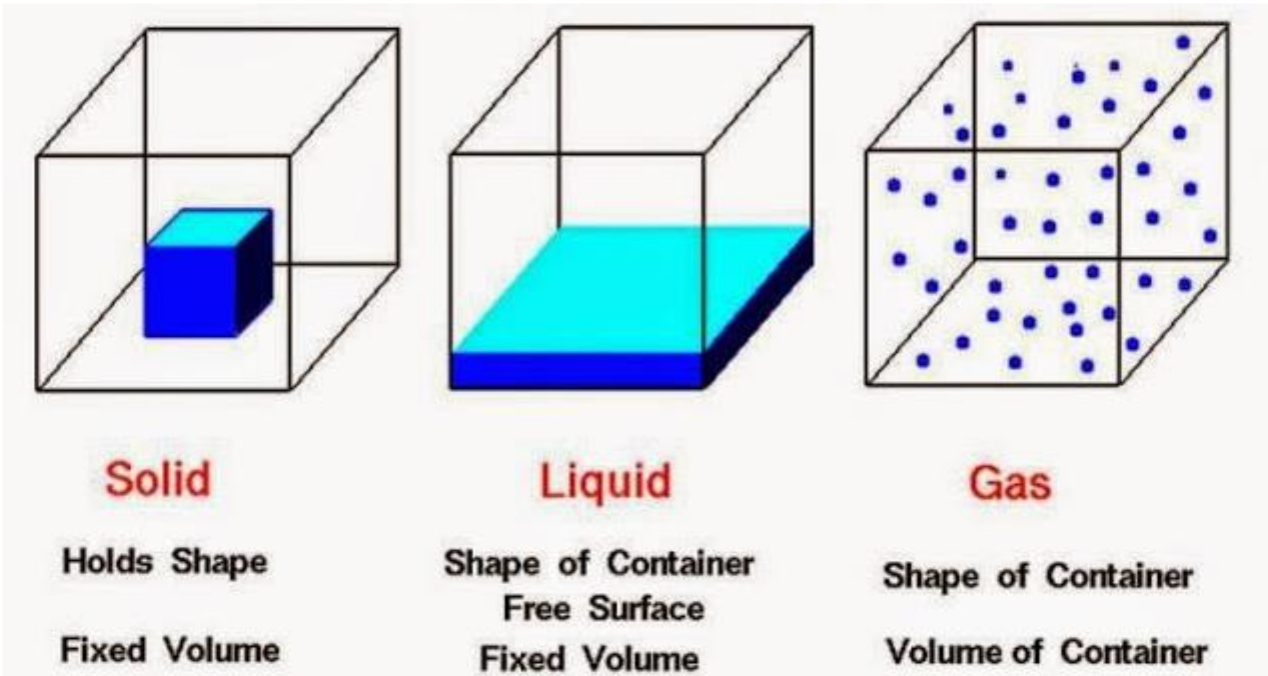
The properties of gases

1.1 The states of gases

What is gas?

- A collection of molecules (or atoms) in continuous random motion
- Speeds increase as the temperature is raised.
- Widely separated from one another and move in paths that are largely unaffected by intermolecular forces




Solid
Holds Shape
Fixed Volume
Liquid
Shape of Container
Free Surface
Fixed Volume
Gas
Shape of Container
Volume of Container

How to describe the physical state of a pure gas?

Physical State of a sample of a substance is defined by its physical properties.

Variables

- Volume (V)
- Pressure (p)
- Amount (n)
- Temperature (T)

Equation of State

$$p = f(T, V, n)$$

Specify 3 variables, the 4th is fixed.

Each substance is described by its own equation of state, but we know the explicit form of the equation in only a few special cases, for example, the perfect gas.

(a) Pressure, *p*

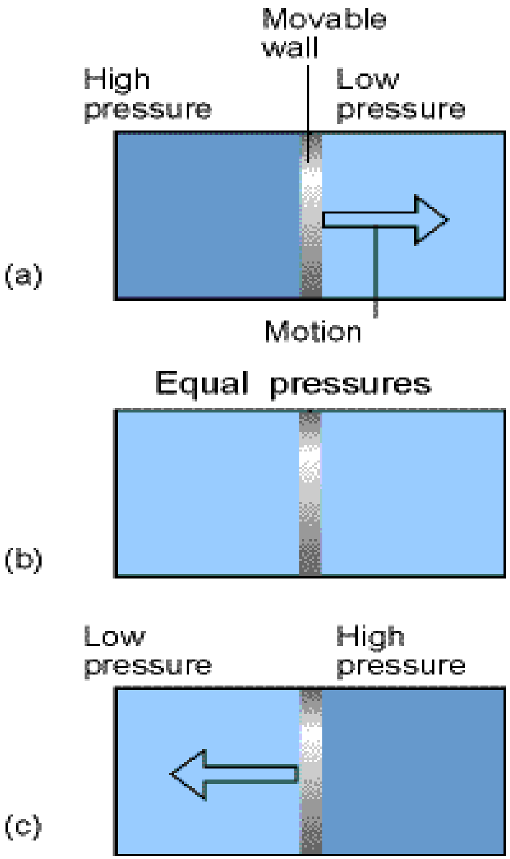
Definition: Pressure is defined as force divided by the area to which the force is applied.

Unit: Pascal (Pa), 1 Pa= 1 Nm^{-2}; the SI unit. It is defined as 1 newton per meter squared;

Standard pressure: $\boldsymbol{p^{\ominus} = 10^{5} Pa} = 1\ \mathrm{bar}$

$1\ atm = 101.325\ kPa$

Mechanical equilibrium



Two gases are in separate containers that share a common movable wall

The stage when the two pressures are equal and the wall has no further tendency to move. This condition of equality of pressure on either side of a movable wall is a state of mechanical equilibrium between the two gases.

$$p_1 = p_2$$

The pressure of a gas is therefore an indication of whether a gas will be in mechanical equilibrium with another gas.

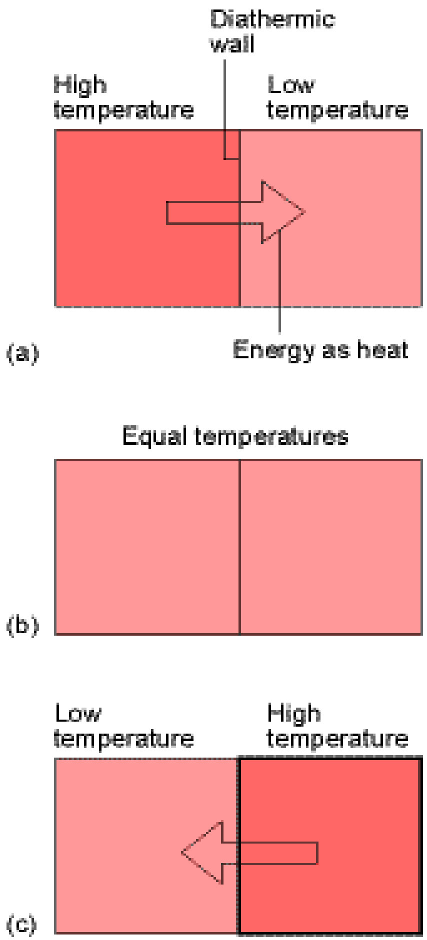
(2) Temperature, T

The temperature is the property that tells us the direction of the flow of energy.

Temperature scales:

- **The Celsius scale of temperature, θ (°C)**
- **The thermodynamic temperature scale, T (K)**

$$T\,/\,\mathrm{K} = \theta\,/\,{}^{\circ}\mathrm{C} + 273.15$$



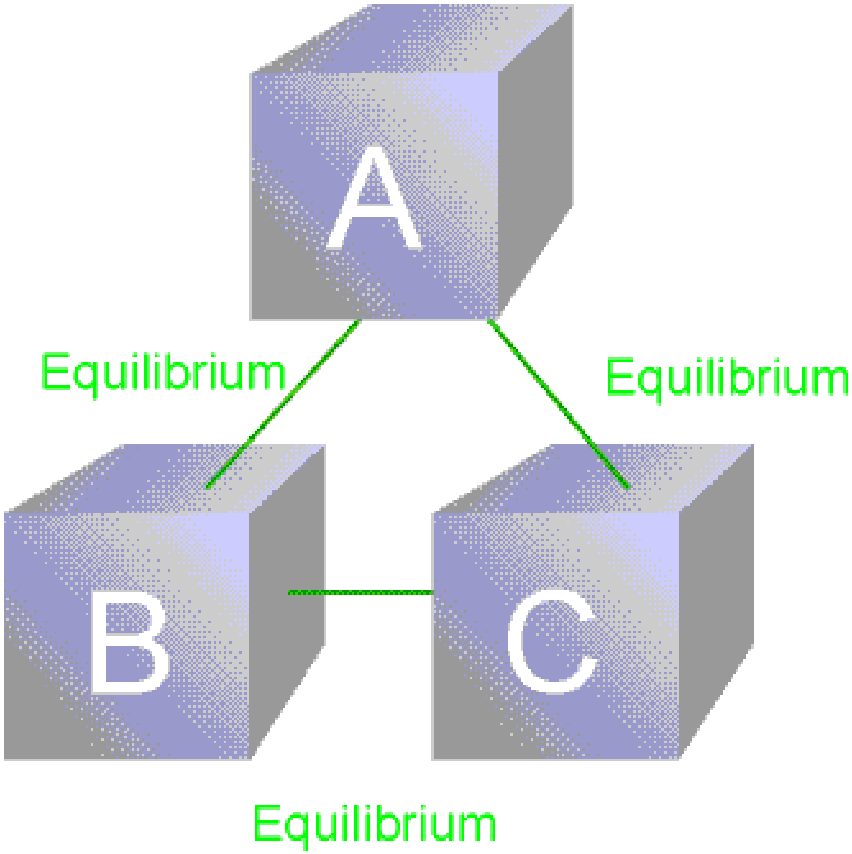
Thermal equilibrium

Energy flows as heat from a region at a higher *T* to one at a lower *T* if the two are in contact through a diathermic wall, as in (a) and (c). However, if the two regions have identical temperatures, there is no net transfer of energy as heat even though the two regions are separated by a diathermic wall (b). The latter condition corresponds to the two regions being at thermal equilibrium: $T_1 = T_2$

Diathermic vs Adiabatic

Zeroth Law of thermodynamics

If an object A is in thermal equilibrium with B and if B is in thermal equilibrium with C, then C is in thermal equilibrium with A.



The Zeroth Law of thermodynamics

(3) Volume, *V*, m^3

(4) The amount of substance

***n*, mole (mol)**

1.2 The gas laws

1.2.1 Boyle's law

n and T are constants

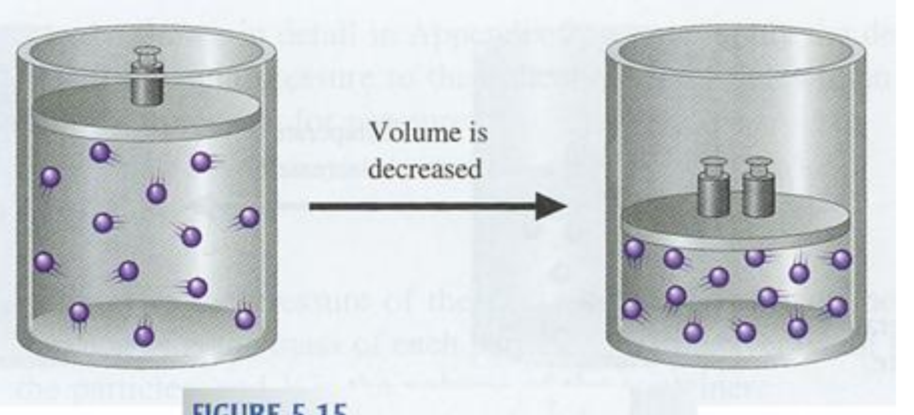
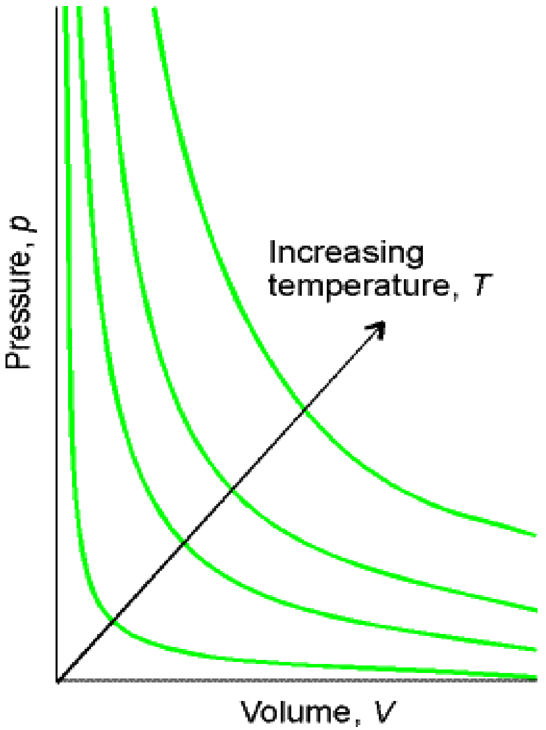


FIGURE 5.15
The effects of decreasing the volume of a sample of gas at constant temperature.

At constant temperature, the pressure of a sample of gas is inversely proportional to its volume, and the volume it occupies is inversely proportional to its pressure.

$$pV = C$$



The *p*-*V* dependence of a fixed amount of perfect gas at different *T*.

Each of the curves in the graph corresponds to a single *T* and is called an isotherm.

1.2.2 Charles's law (Gay-Lussac's law)

At constant pressure, the volume of the gas increased linearly with the temperature.

$V = CT$ *(at constant pressure)*

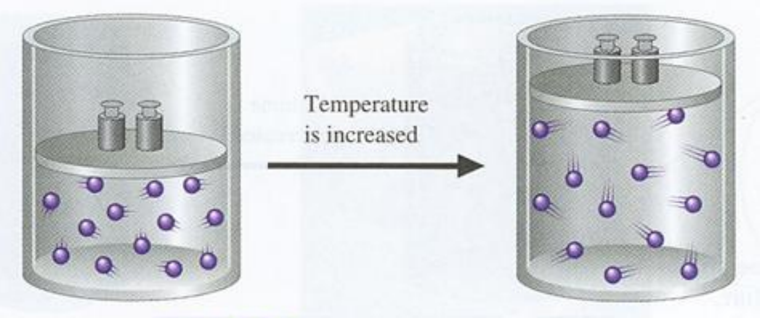


FIGURE 5.17
The effects of increasing the temperature of a sample of gas at constant pressure.

At constant volume, the pressure of the gas increased linearly with the temperature.

$p = CT$ *(at constant volume)*

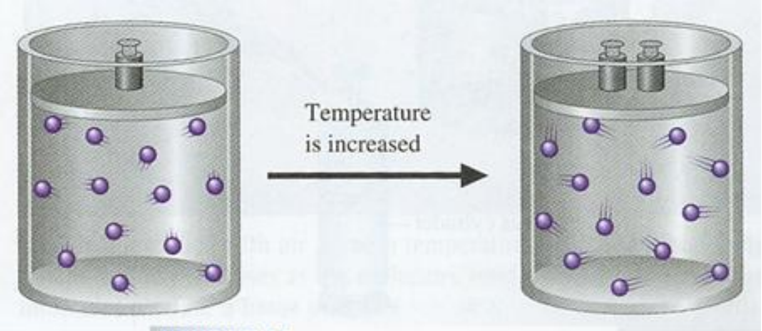


FIGURE 5.16
The effects of increasing the temperature of a sample of gas at constant volume.

The limiting law

Boyle's and Charles's laws are examples of a limiting law, a law that is strictly true only in a certain limit, in this case $p \rightarrow 0$; and that real gases obey these laws only in the limit of the pressure approaching zero.

Equations that are valid in this limiting sense will be signaled by a ° on the equation number in our text book.

1.2.3 Avogadro's principle

at constant T and p:

$$V = \text{constant} \times n$$

$$V_m = {}^{V}/_{n} = constant$$

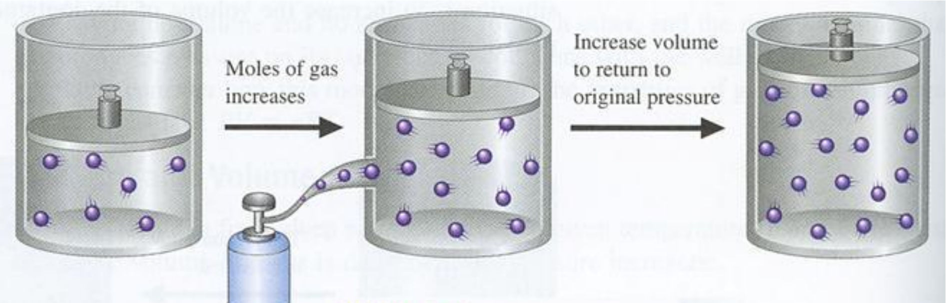
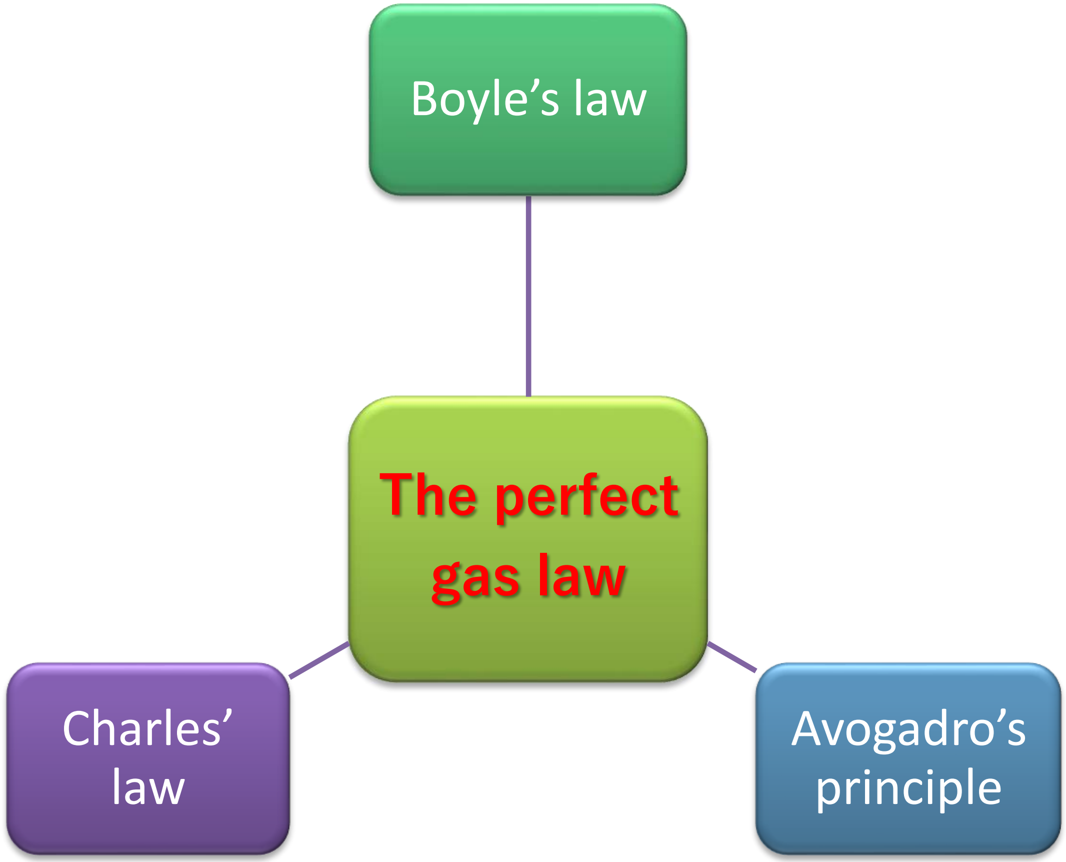


FIGURE 5.18 The effects of increasing the number of moles of gas particles at constant temperature and pressure.

At a given pressure and temperature, the molar volume of a gas is approximately the same; the volume of a sample of gas is proportional to the amount of molecules present.

Avogadro's principle: The equal volumes of gases at the same pressure and temperature contain the same numbers of molecules.


Boyle's law
The perfect gas law
Charles' law
Avogadro's principle

1.2.4 The perfect gas law

$$pV = constant \times nT$$

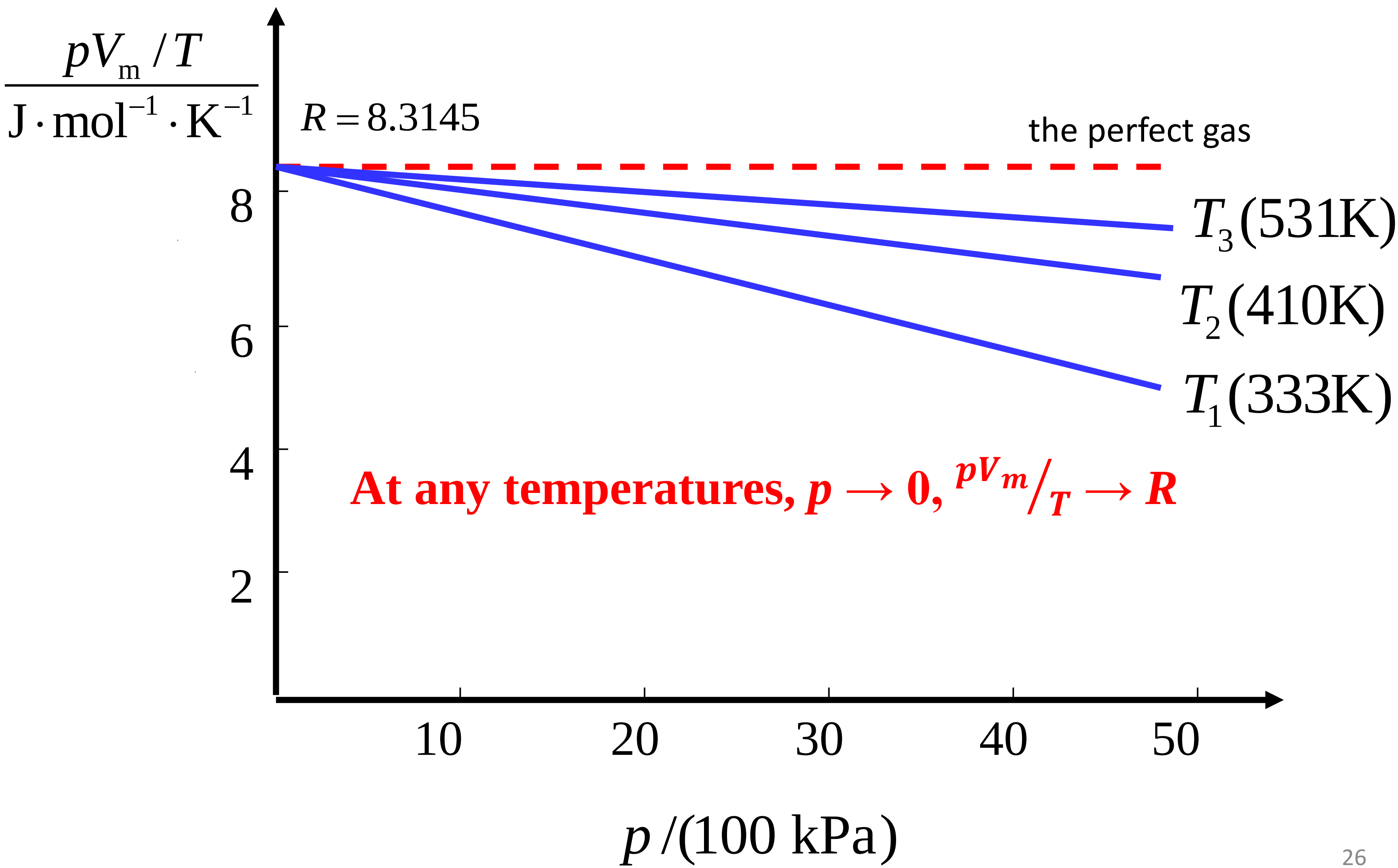
- A gas that obeys this Equation exactly under all conditions is called a perfect gas.
- A real gas behaves more like a perfect gas the lower the pressure, and is described exactly by this equation in the limit $p \to 0$.

$$pV = nRT$$

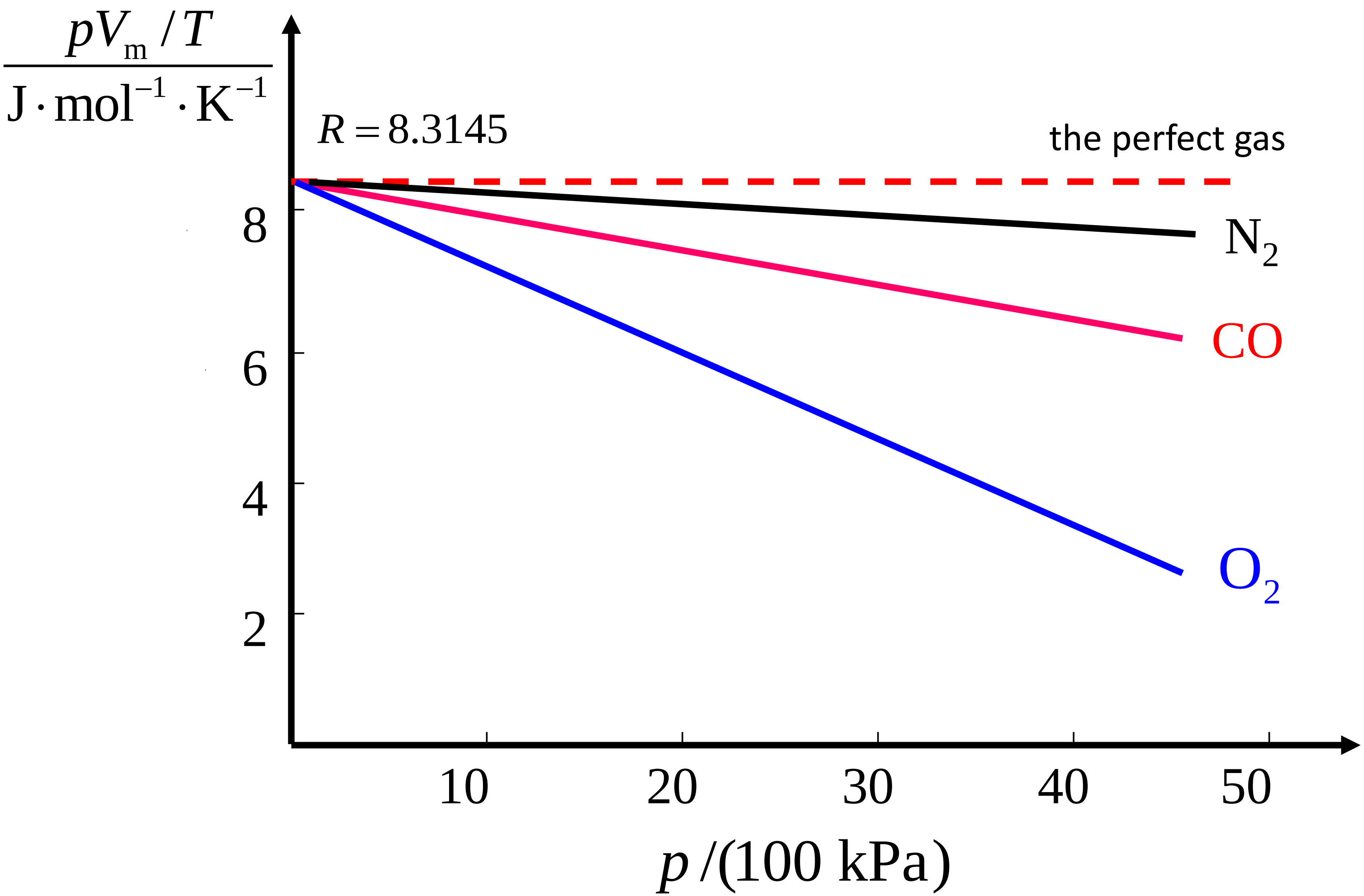
R: gas constant

$$p\frac{V}{n} = RT \quad \Rightarrow \quad R = {pV_m}/{T}$$

V_m: *molar volume*



At any temperatures, $p \to 0$, $pV_m/T \to R$



$\dfrac{pV_m/T}{J \cdot mol^{-1} \cdot K^{-1}}$
$R = 8.3145$
the perfect gas
8
6
4
2
N_2
CO
O_2
10
20
30
40
50
$p/(100\ kPa)$

$$R = \frac{1\text{atm} \times 22.414\text{L}}{1\text{mol} \times 273.15\text{K}} = 8.20578 \times 10^{-2}\ \text{L atm mol}^{-1}\ \text{K}^{-1}$$

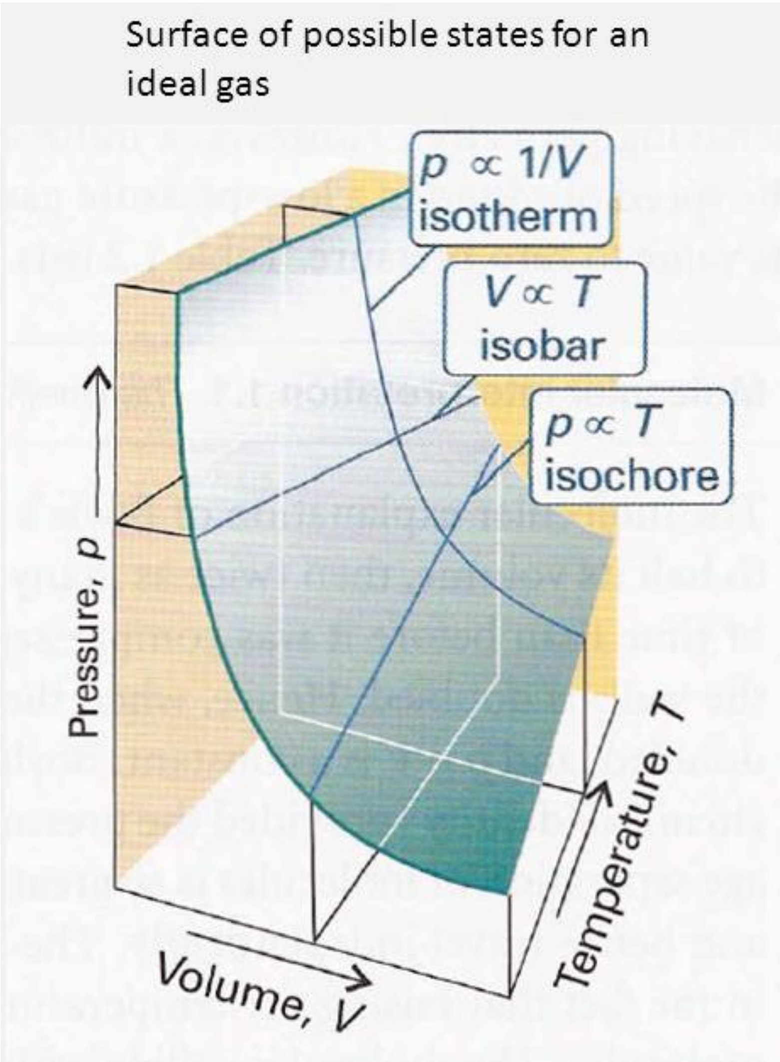
$$= 8.314\text{J mol}^{-1}\ \text{K}^{-1}$$

TABLE 5.2 Molar Volumes for Various Gases at 0°C and 1 atm

Gas	Molar Volume (L)
Oxygen (O_2)	22.397
Nitrogen (N_2)	22.402
Hydrogen (H_2)	22.433
Helium (He)	22.434
Argon (Ar)	22.397
Carbon dioxide (CO_2)	22.260
Ammonia (NH_3)	22.079

R with different units

8.31451	J K^{-1} mol^{-1}
8.20578 x 10^{-2}	L atm K^{-1} mol^{-1}
8.31451 x 10^{-2}	L bar K^{-1} mol^{-1}
8.31451	Pa m^3 K^{-1} mol^{-1}
62.364	L Torr K^{-1} mol^{-1}
1.98722	cal K^{-1} mol^{-1}



The combined gas equation

$$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$$

Exercise 1:

In an industrial process, nitrogen is heated to 500 K in a vessel of constant volume. If it enters the vessel at 100 atm and 300 K, what pressure would it exert at the working temperature if it behaved as a perfect gas?

Solution:

	n	p	V	T
1	same	100	same	300
2	same	?	same	500

$$\frac{p_1}{T_1} = \frac{p_2}{T_2} \longrightarrow p_2 = \frac{T_2}{T_1} \times p_1 = \frac{500K}{300K} \times 100\ atm = 167\ atm$$

1.2.5 Dalton’s law- mixtures of gases

gas mixture	gas A	gas B
$n=n_A+n_B$	n_A	n_B
$P \quad V$	$P_A \quad V$	$P_B \quad V$

$$p=(n_A+n_B)\frac{RT}{V}=n_A\frac{RT}{V}+n_B\frac{RT}{V}=p_A+p_B$$

partial pressure

$$p_i=n_i\frac{RT}{V}$$

The partial pressure of a perfect gas is the pressure that it would exert if it occupied the container alone.

Mole fractions and partial pressures

Mole fraction x_i

The amount of *i* expressed as a fraction of the total amount of molecules n.

$$x_i=\frac{n_i}{n} \qquad n=n_A+n_B+\ldots$$

$$x_A+x_B+\ldots=1$$

Partial pressure p_i

A gas i in a mixture

$$p=(n_A+n_B+\cdots)\frac{RT}{V} \qquad p_i=n_i\frac{RT}{V}$$

$$p_i=x_ip \qquad p_A+p_B+\cdots=(x_A+x_B+\cdots)p=p$$

Dalton's Law

The pressure exerted by a mixture of gases is the sum of the pressures that each one would exist if it occupied the container alone.

$$p_A + p_B + \cdots = p$$ ⇨ For all gases

$$p_i = n_i \frac{RT}{V}$$ ⇨ For perfect gases

Exercise 2:

A container of volume 10.0 L holds 1.00 mol N_2 and 3.00 mol H_2 at 298 K. What is the total pressure in atmospheres if each component behaves as a perfect gas?

Since

$$p_i = n_i \frac{RT}{V}$$

We have

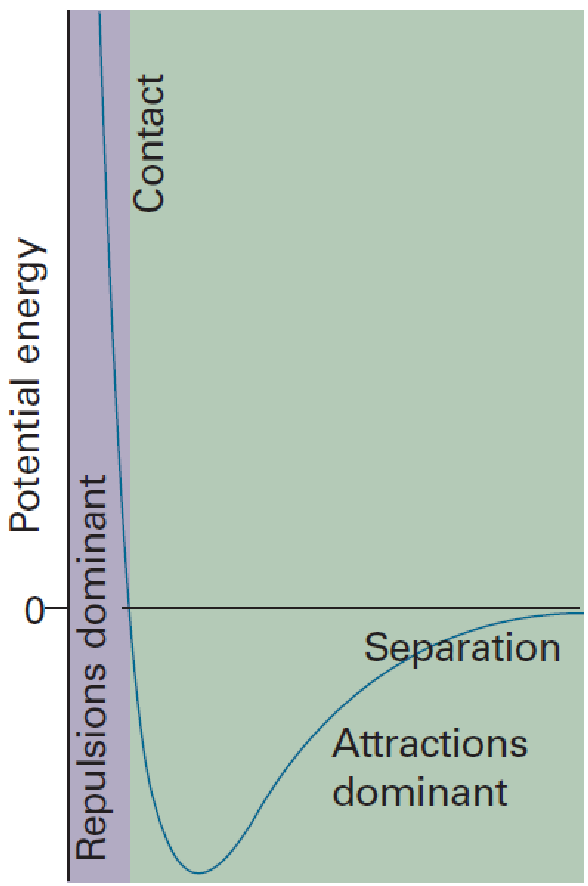
$$p = p_{\mathrm{A}} + p_{\mathrm{B}} = (n_{\mathrm{A}} + n_{\mathrm{B}}) \frac{RT}{V}$$

$$= (1.00\mathrm{mol} + 3.00\mathrm{mol}) \times \frac{(8.206 \times 10^{-2}\,\mathrm{LatmK^{-1}mol^{-1}}) \times 298\mathrm{K}}{10.0\mathrm{L}}$$

$$= 9.87\mathrm{atm}$$

1.3 Real gas

1.3.1 Intermolecular interaction



- Repulsive forces are significant only when molecules are almost in contact.
- Attractive intermolecular forces have a relatively long range and are effective over several molecular diameters. Attractive forces are ineffective when the molecules are far apart.
- Intermolecular forces are also important when the temperature is so low that the molecules travel with such low mean speeds that they can be captured by one another.

1.3.2 The compression factor: Z

- Real gases show deviations from the perfect gas law because molecules interact with each other.
- Repulsive forces between molecules assist expansion.
- Attractive forces assist compression.

Z of a gas is the ratio of its molar volume, V_m, to the molar volume of a perfect gas, V_m°, at the same pressure and temperature.

$$Z = \frac{V_m}{V_m^{\circ}}$$

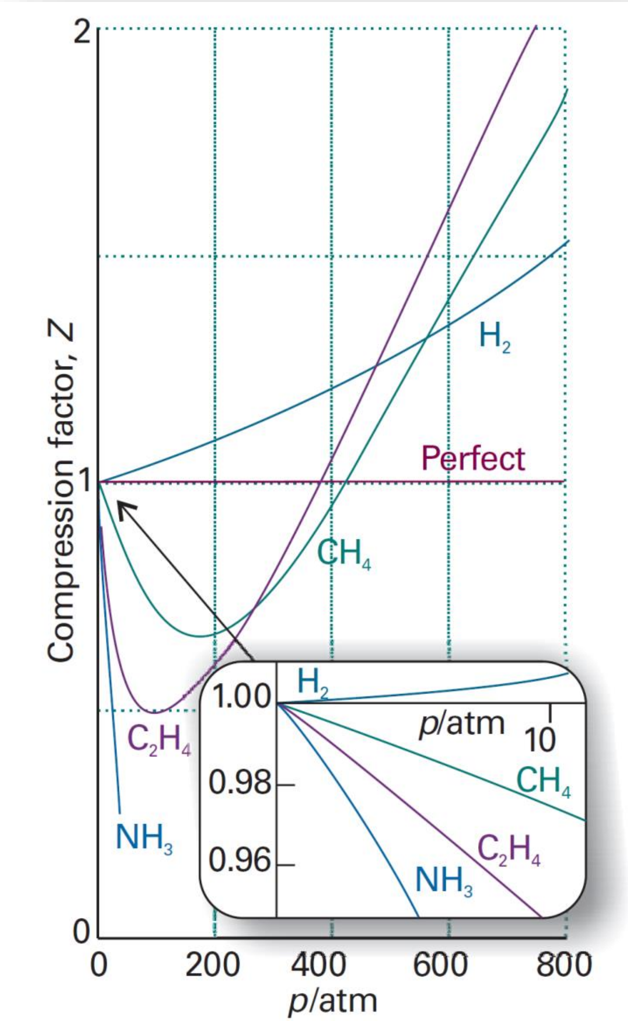
$$pV_m^{\circ} = RT \Rightarrow pV_m = ZRT$$

The perfect gases $Z = \frac{pV_m}{RT} = 1$

Real gases

$Z > 1$ $V_m > V_m^{\circ}$ expansion

$Z < 1$ $V_m < V_m^{\circ}$ compression



Results by plotting the compression factor, Z, against pressure for several gases at 0 ℃.

At very low pressures: Z ≈ 1.

At high pressures: Z> 1.

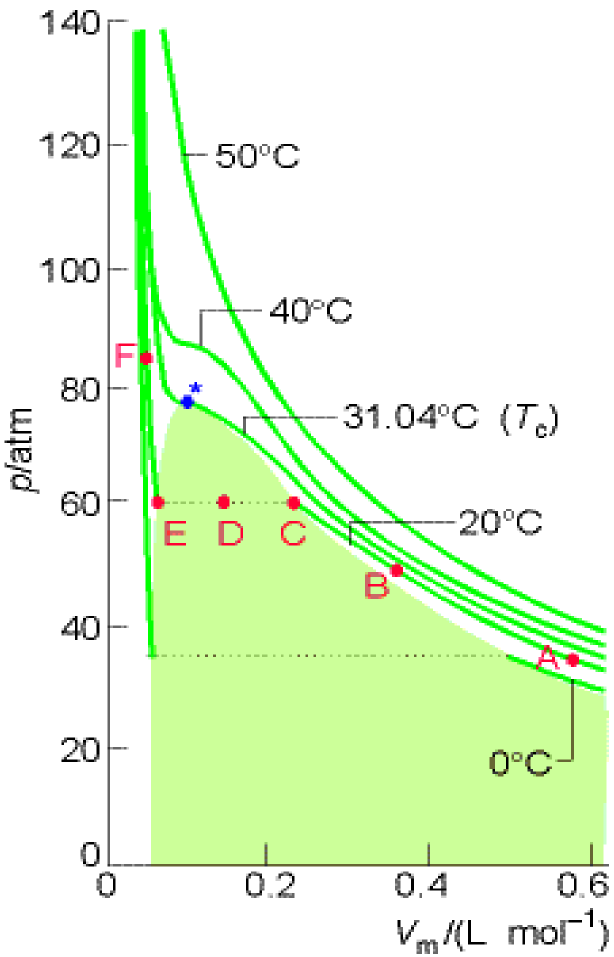
Repulsive forces are now dominant.

At intermediate pressures: Z<1

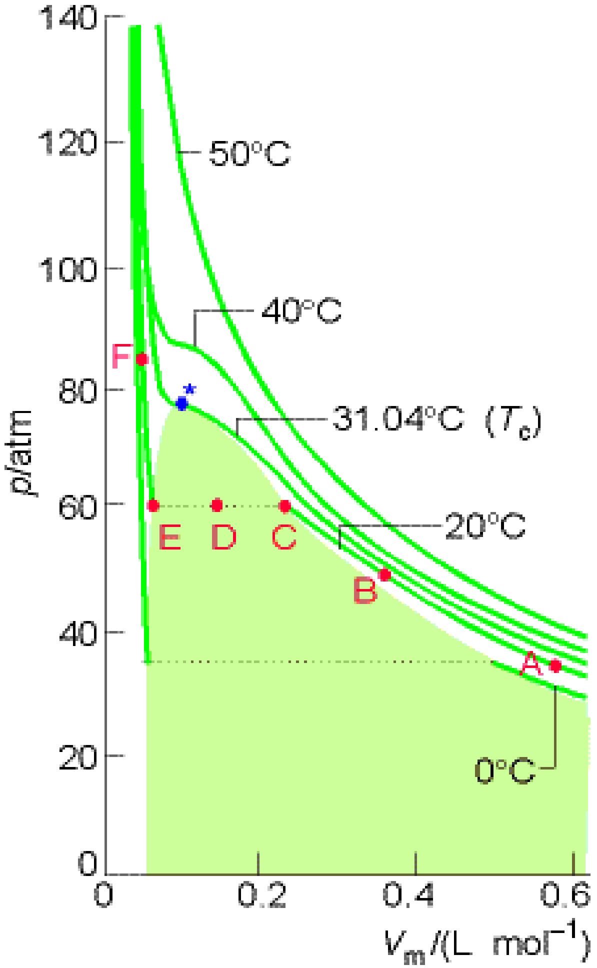
For most gases, the attractive forces are dominant.

A perfect gas has Z = 1 at all pressures.

1.3.3 Virial coefficients（an empirical method）



At large molar volumes and high temperatures the real isotherms do not differ greatly from perfect isotherms. The small differences suggest that the perfect gas law is in fact the first term in the expression of the form:



These are two versions of the virial equation of state:

$$pV_m = RT(1 + B'p + C'p^2 + \cdots)$$

$$pV_m = RT\left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots\right)$$

The coefficients *B*, *C*,..., which depend on the *T*, are the second, third,... virial coefficients; the first virial coefficient is 1.

We can use the virial equation to demonstrate the important point that, although the equation of state of a real gas may coincide with the perfect gas law as $p \to 0$, not all its properties necessarily coincide with those of a perfect gas in that limit.

$$pV_m = RT(1 + B'p + C'p^2 + \cdots)$$

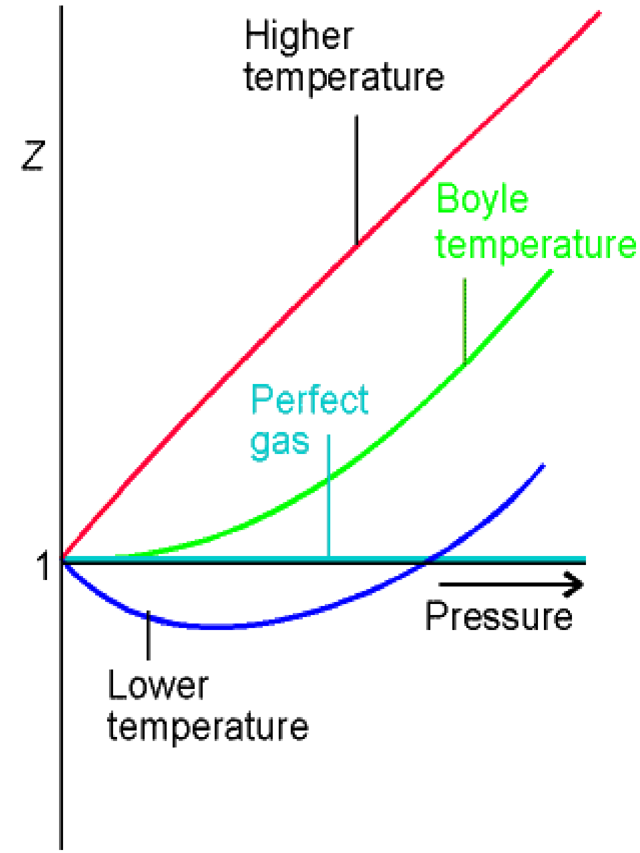
$$pV_m = ZRT$$

$$Z = 1 + B'p + C'p^2 + \cdots$$

$$\left(\frac{\partial Z}{\partial p}\right)_T = B' + 2C'p + \cdots \to B' \quad \text{as} \quad p \to 0$$

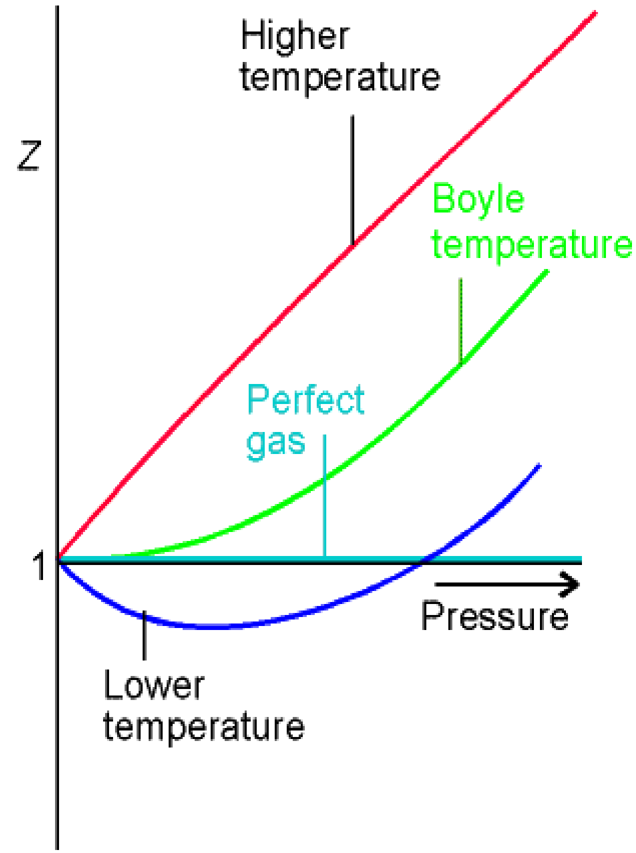
$$\left(\frac{\partial Z}{\partial p}\right)_{T,p\to 0} = B' > 0 \quad at\ higher\ temeprature$$

$$\left(\frac{\partial Z}{\partial p}\right)_{T,p\to 0} = B' < 0 \quad at\ lower\ temeprature$$



$$\left(\frac{\partial Z}{\partial p}\right)_{T,p\rightarrow 0} = B' = 0 \quad at\ Boyle\ temperature\ T_B$$

At the **Boyle temperature**, T_B, the properties of the real gas do coincide with those of a perfect gas as $p \rightarrow 0$.



1.3.4 The van der Waals equation（理论推理）

For the perfect gas:

1). The size of molecules is negligible;

2). Molecules do not interact;

…

$$pV=nRT$$

For real gases:

1). The molecule itself occupies a volume;

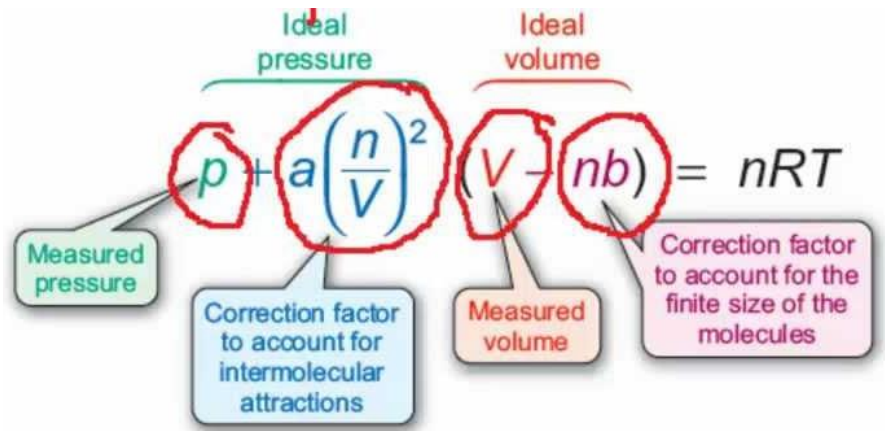
2). There are interactions among molecules;

…

$$(p+\triangle p)(V-\triangle V)=nRT$$

$$(p+\triangle p)\ (V-\triangle V\ )=nRT$$

$$\left(p+\frac{n^2a}{V^2}\right)(V-nb)=nRT$$



van der Waals equation

$$\left(p+\frac{a}{V_m{}^2}\right)(V_m-b)=RT \Rightarrow p=\frac{RT}{V_\mathrm{m}-b}-\frac{a}{V_\mathrm{m}^2}$$

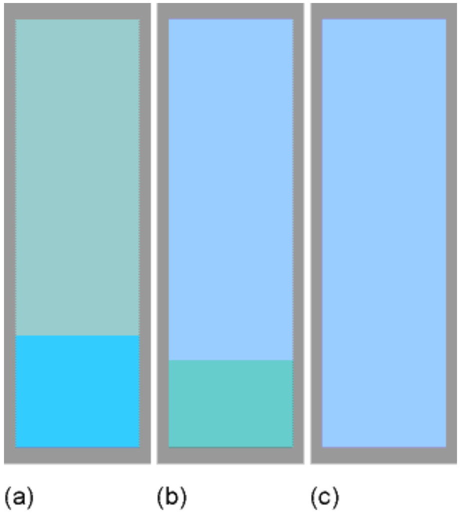
The constants a and b are the van der Waals coefficients. They are characteristic of each gas but independent of the temperature.

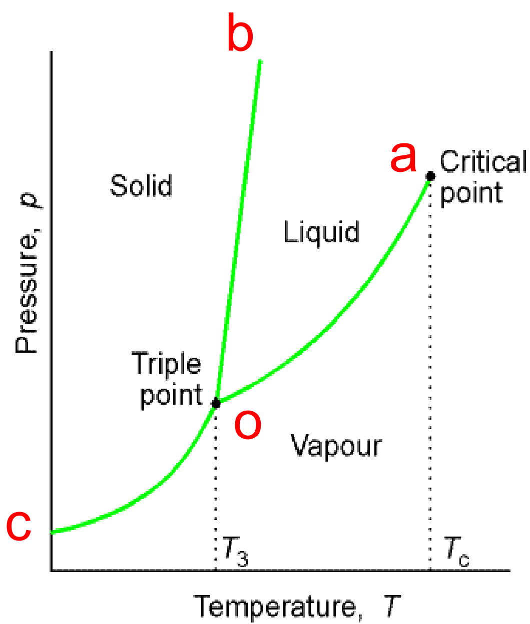
- a represents the strength of attractive interactions and b that of the repulsive interactions between the molecules.

- Although a and b are not precisely defined molecular properties, they correlate with physical properties such as critical temperature, vapour pressure, and enthalpy of vaporization that reflect the strength of intermolecular interactions.

- We still expect a single, simple expression to be the true equation of state of all substances.

Critical constants





<u>In a closed vessel</u>

(a) In equilibrium (vapour pressure).

(b) With heating, the density of the vapour phase increases and that of the liquid decreases slightly.

(c)There comes a stage, at which the two densities are equal and the l-g interface disappears.

Critical temperature (T_c)

The temperature at which the surface disappears.

Critical pressure (p_c)

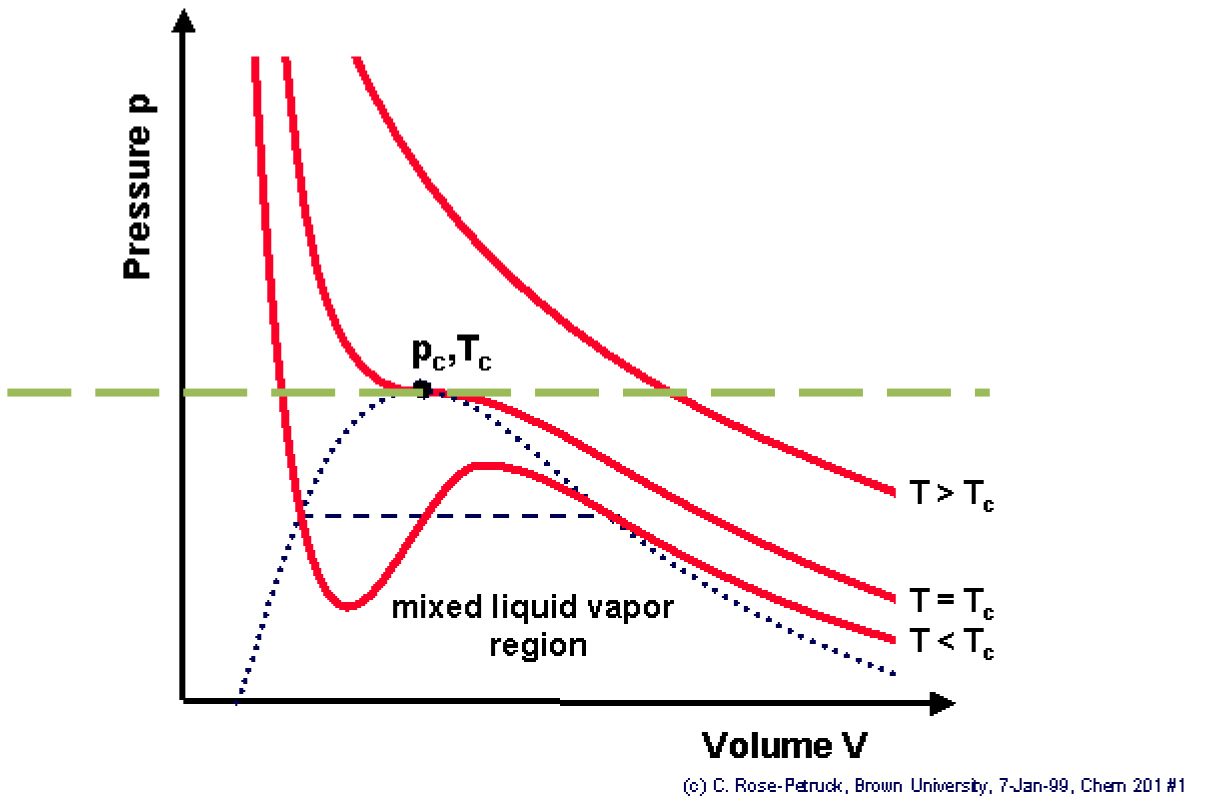
The vapour pressure at T_c

Supercritical fluid

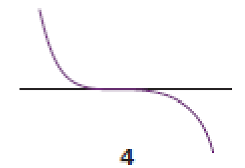
The single uniform phase at T_c, p_c.

T_c and p_c are characteristics of a substance.

B. Van der Waals isotherms



Critical point



$$\left(\frac{\partial p}{\partial V}\right)_{T_\mathrm{c}} = 0 \qquad \left(\frac{\partial^2 p}{\partial V^2}\right)_{T_\mathrm{c}} = 0 \qquad p = \frac{RT}{V_\mathrm{m} - b} - \frac{a}{V_\mathrm{m}^2}$$

$$\left(\frac{\partial p}{\partial V_\mathrm{m}}\right)_{T_\mathrm{c}} = \frac{-RT_\mathrm{c}}{(V_\mathrm{m} - b)^2} + \frac{2a}{V_\mathrm{m}^3} = 0 \qquad \left(\frac{\partial^2 p}{\partial V_\mathrm{m}^2}\right)_{T_\mathrm{c}} = \frac{2RT_\mathrm{c}}{(V_\mathrm{m} - b)^3} - \frac{6a}{V_\mathrm{m}^4} = 0$$

$$V_\mathrm{m,c} = 3b \qquad T_\mathrm{c} = \frac{8a}{27Rb} \qquad p_\mathrm{c} = \frac{a}{27b^2} \qquad R = \frac{8}{3}\frac{p_\mathrm{c} V_\mathrm{m,c}}{T_\mathrm{c}}$$

$$a = \frac{27}{64}\frac{R^2 T_\mathrm{c}^2}{p_\mathrm{c}} \qquad b = \frac{RT_\mathrm{c}}{8p_\mathrm{c}}$$

critical compression factor $Z_c = \frac{p_c V_{m,c}}{RT_c} = \frac{3}{8}$

C. The principle of corresponding states对比状态定律

$$a=\frac{27}{64}\frac{R^2T_c^2}{p_c} \quad b=\frac{RT_c}{8p_c} \quad R=\frac{8}{3}\frac{p_cV_{m,c}}{T_c}$$

the van der Waals equation

$$\left(\frac{p}{p_c}+\frac{3V_{m,c}^{2}}{V_m^{2}}\right)\left(\frac{V_m}{V_{m,c}}-\frac{1}{3}\right)=\frac{8}{3}\frac{T}{T_c}$$

Reduced variables (对比状态):

Dividing the actual variable by the corresponding critical constant

$$p_r=\frac{p}{p_c}\quad,\quad V_r=\frac{V_m}{V_{m,c}}\quad,\quad T_r=\frac{T}{T_c}$$

Real gases at the same reduced volume and reduced temperature exert the same reduced pressure

$$\left(p_r+\frac{3}{V_r^{2}}\right)\left(V_r-\frac{1}{3}\right)=\frac{8}{3}T_r$$

D. The compressibility factor chart

$$Z=\frac{pV_m}{RT}=\frac{p_cV_{m,c}}{RT_c}\cdot\frac{p_rV_r}{T_r}=Z_c\frac{p_rV_r}{T_r}$$

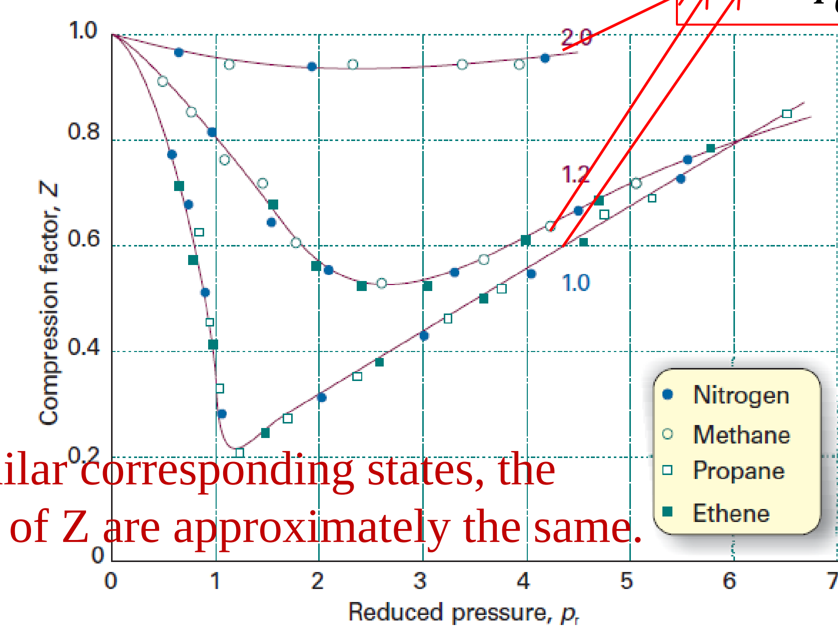
$$T_r=\frac{T}{T_c}$$

$$Z_c=\frac{p_cV_{m,c}}{RT_c}=\frac{3}{8}$$

$Z_c \approx 0.375$

If p_r , V_r , T_r are the same, Z is the same

$Z = f\,(T_r\,,\,p_r\,)$ —— for all real gases



At similar corresponding states, the values of Z are approximately the same.

Fig. 1.19 The compression factors of four of the gases shown in Fig. 1.14 plotted using reduced variables. The curves are labelled with the reduced temperature $T_r = T/T_c$. The use of reduced variables organizes the data on to single curves.

对比状态原理是van der waals方程最有用的副产品，在工程上有广泛应用

Summary

1. The perfect gas equation
2. The Dalton’s law
3. Compression factor
4. Virial coefficient
5. The van der Waal’s equation
6. The principle of corresponding states
7. Use compressibility factor chart to calculate the state of real gases